Support Vector Machines

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Topics

- Maximum Margin Classifiers
- Soft Margin Classifiers
- Support Vector Classifiers
- Support Vector Machines
- Math behind SVM and its explanation
- Kernel Trick
- Support Vector Regression
- Support Vector Clustering
- One Class SVM

Key Concepts

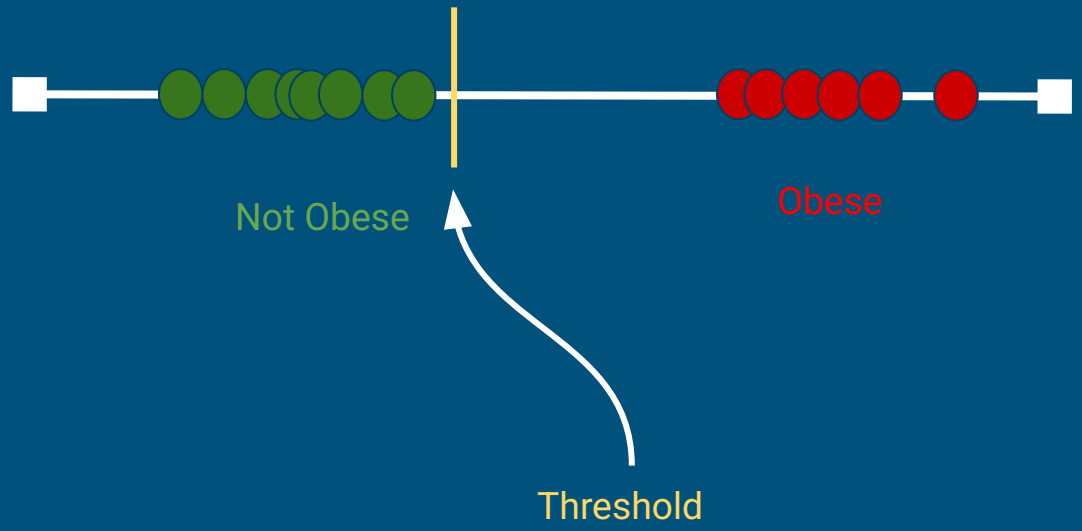
Maximum Margin Classifiers

Maximum Margin Classifiers

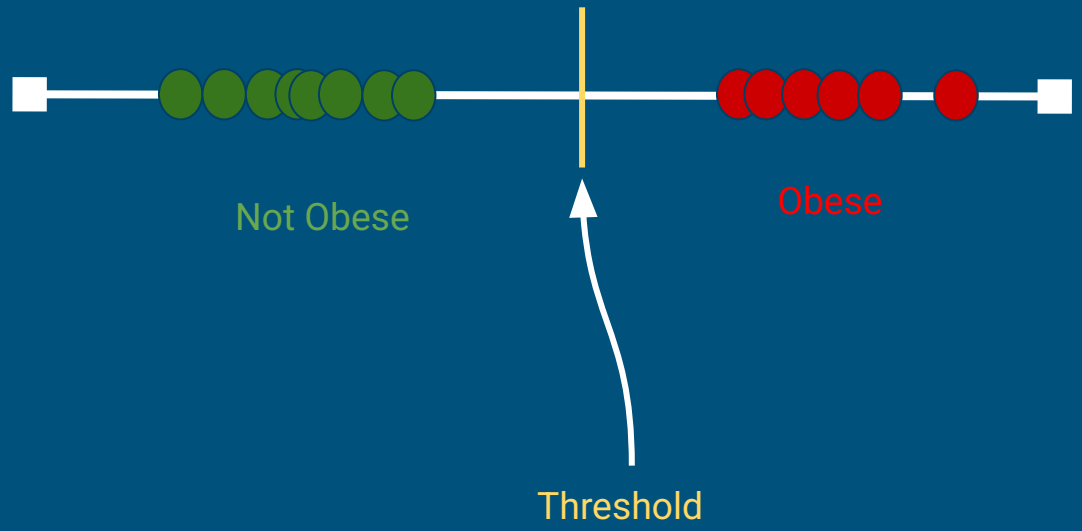
BMI data

Maximum Margin Classifiers

Maximum Margin Classifiers



Maximum Margin Classifiers



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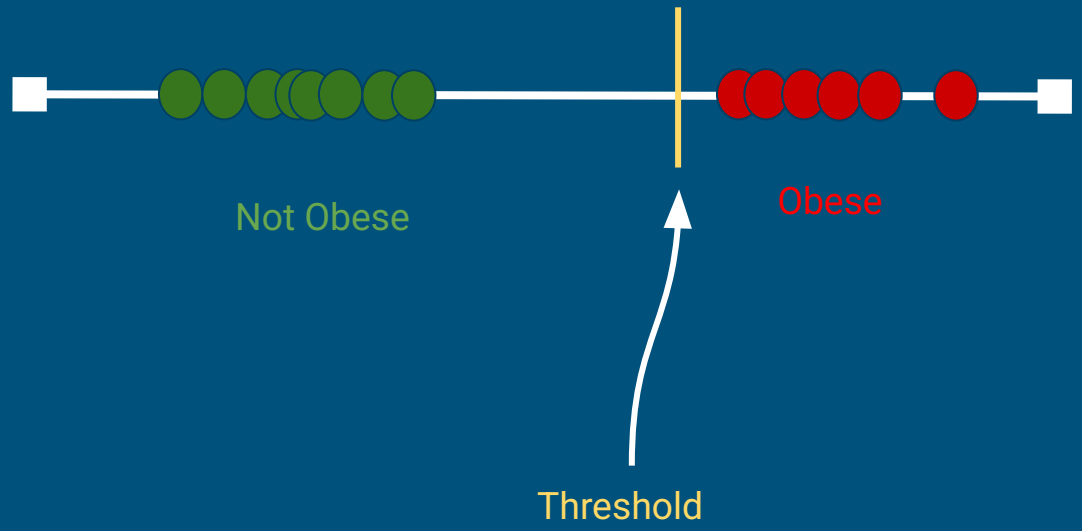

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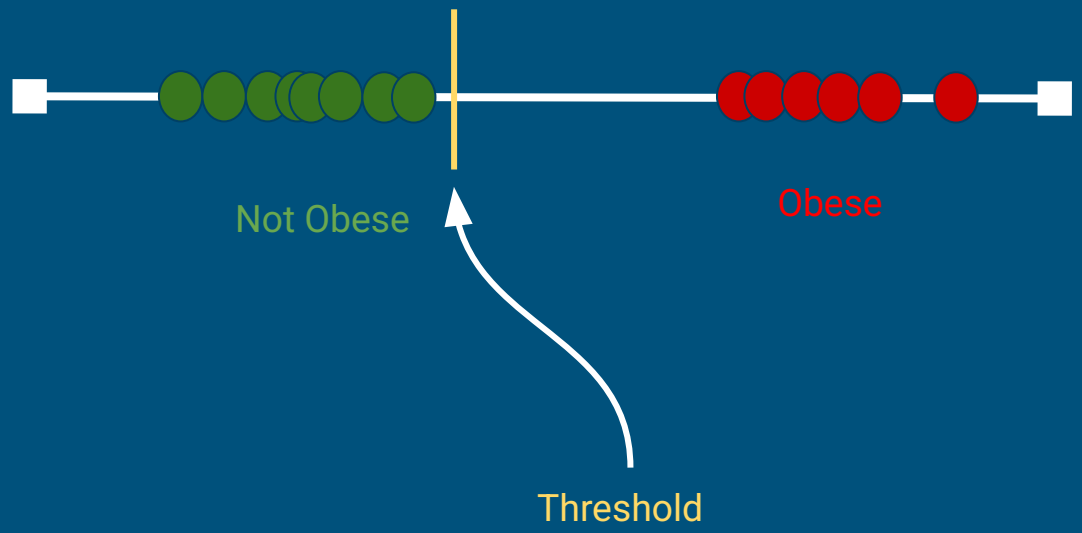
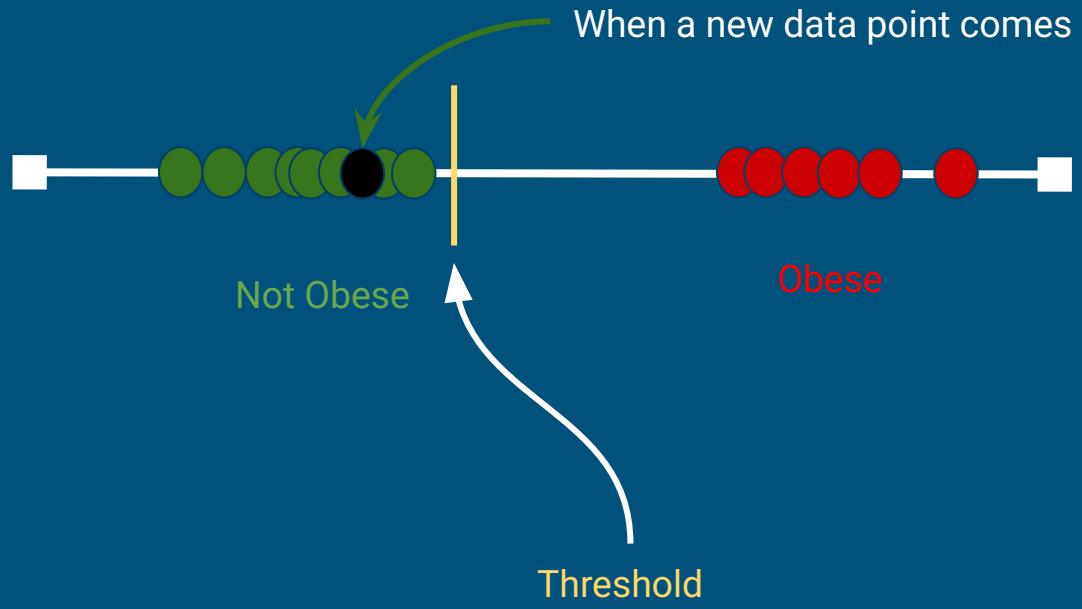
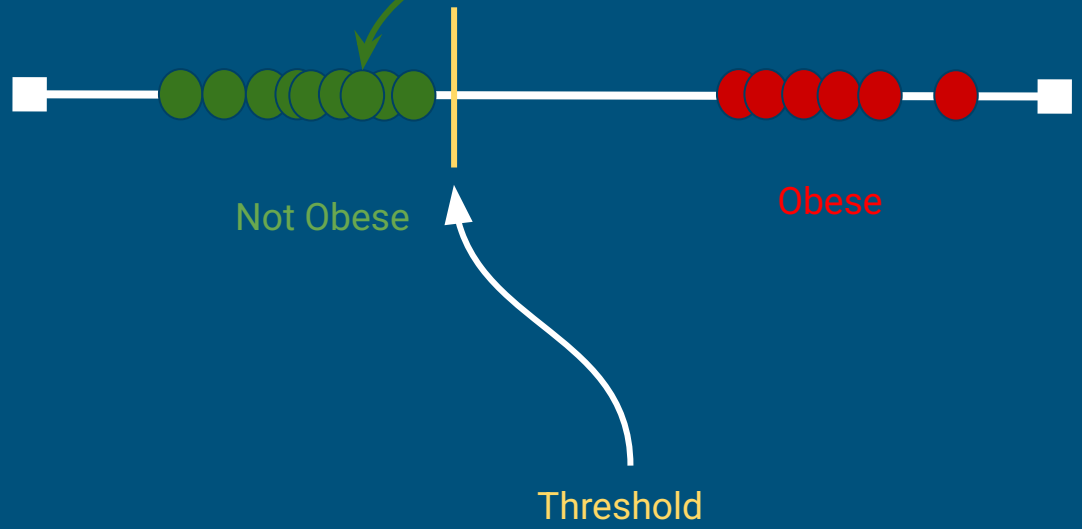


Introduction



Maximum Margin Classifiers

When a new data point comes - we classify it as ***not obese***



Maximum Margin Classifiers

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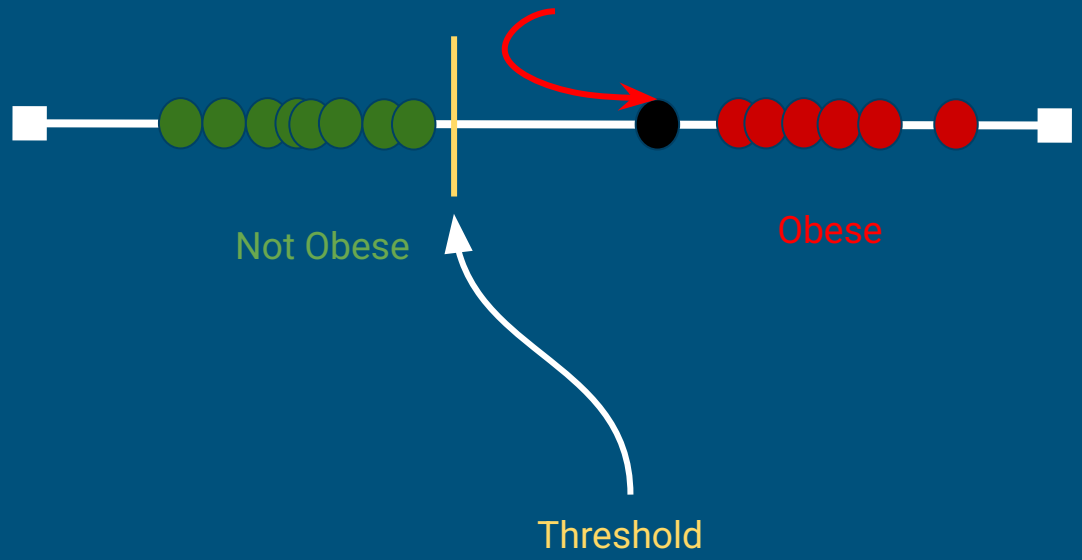
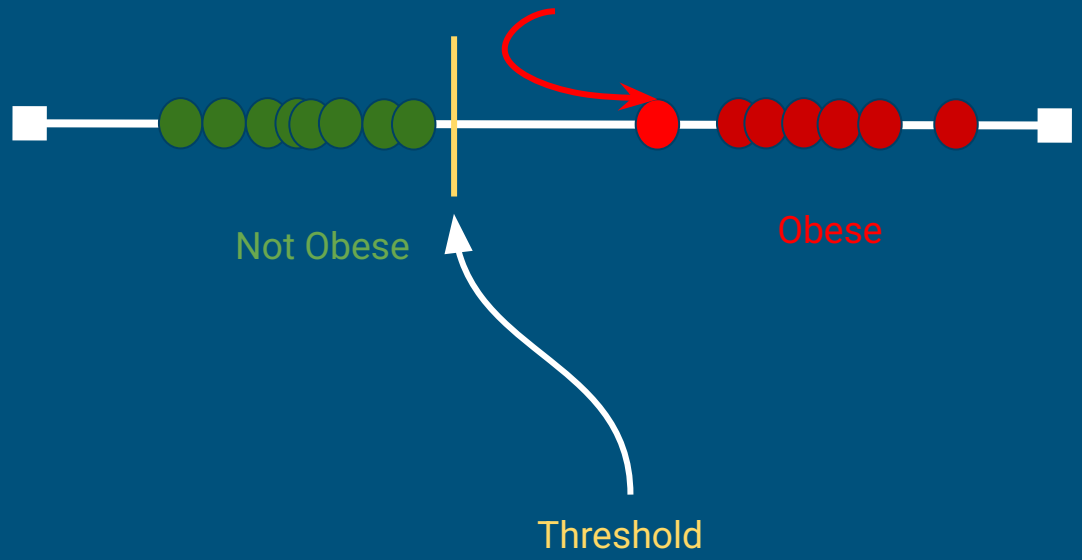
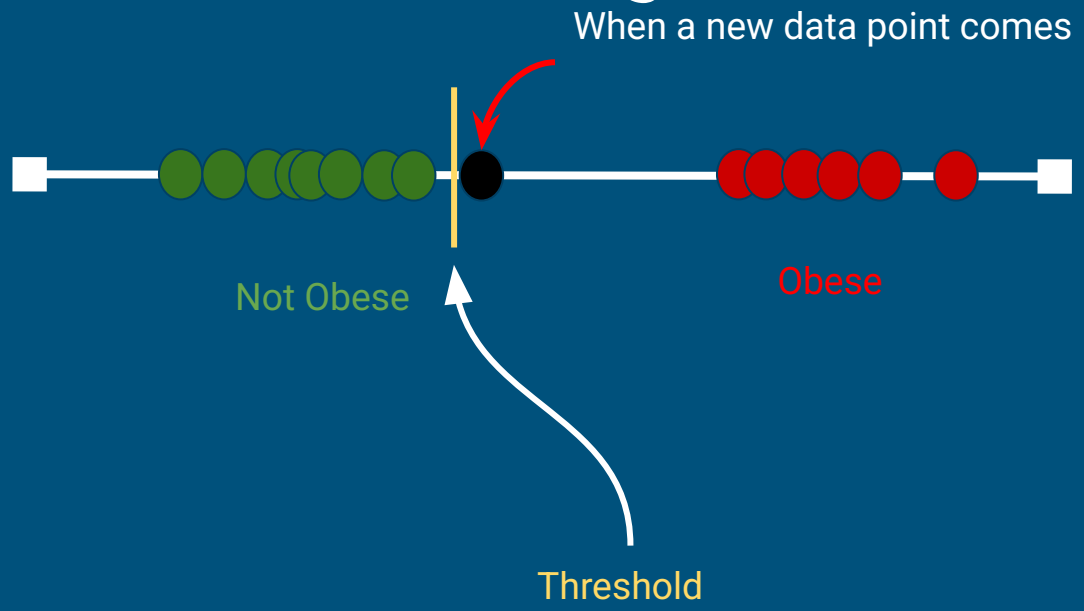


Maximum Margin Classifiers

When a new data point comes - we classify it as ***obese***



Maximum Margin Classifiers



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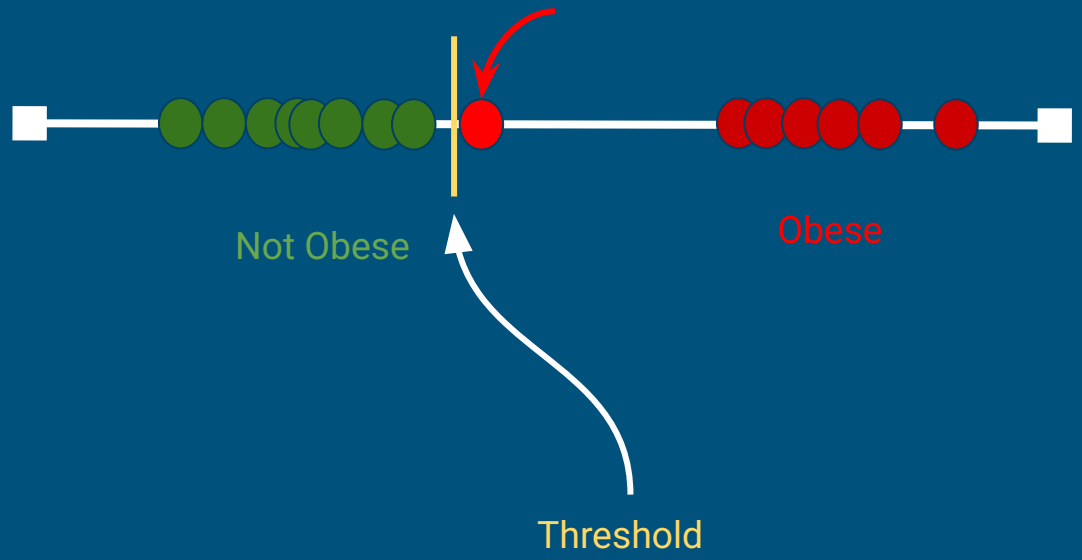

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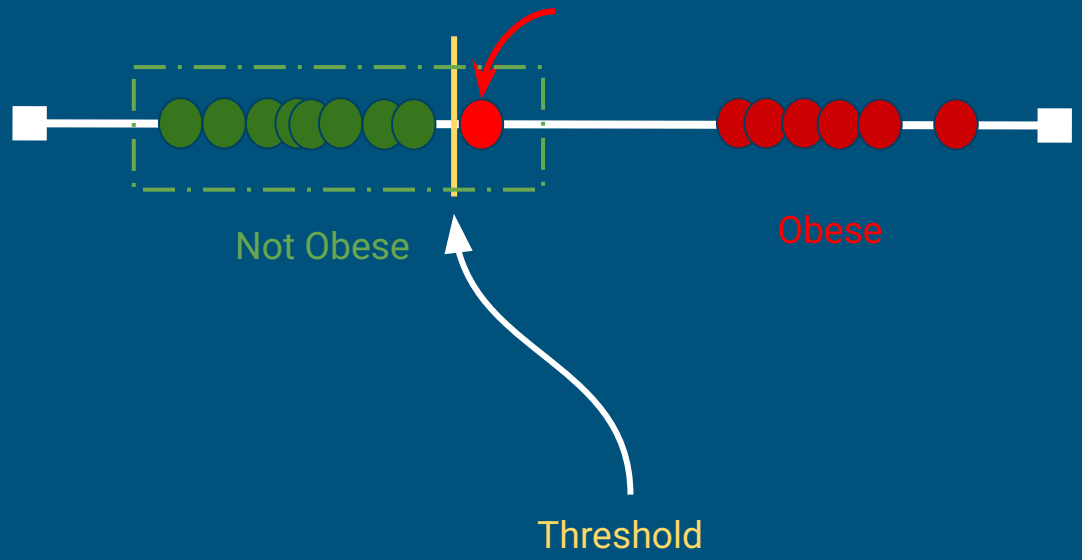

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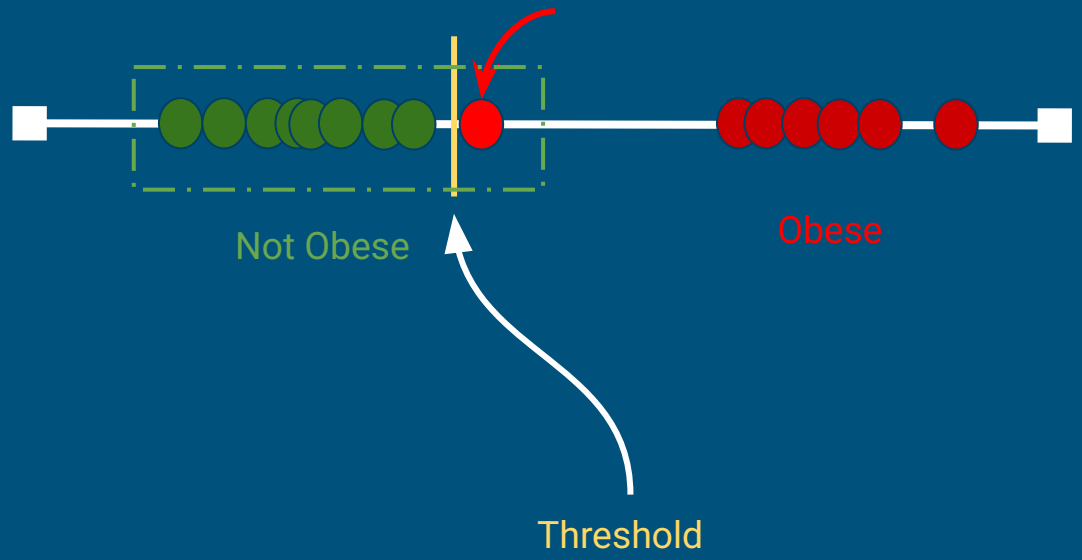

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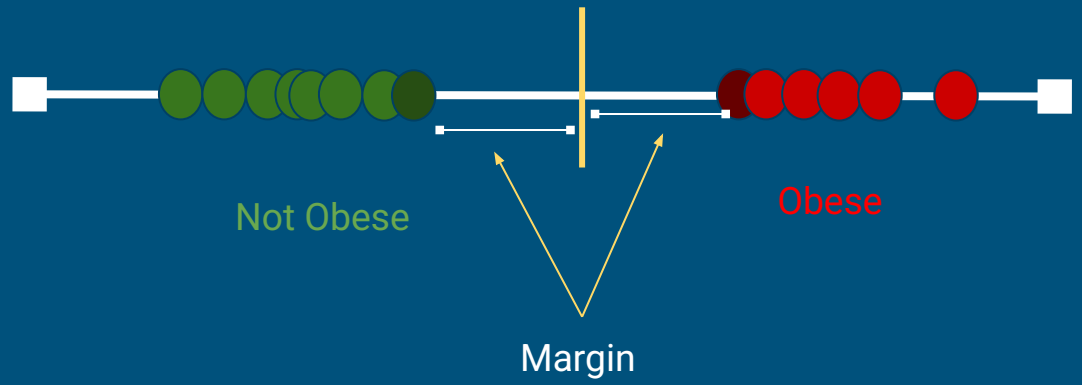
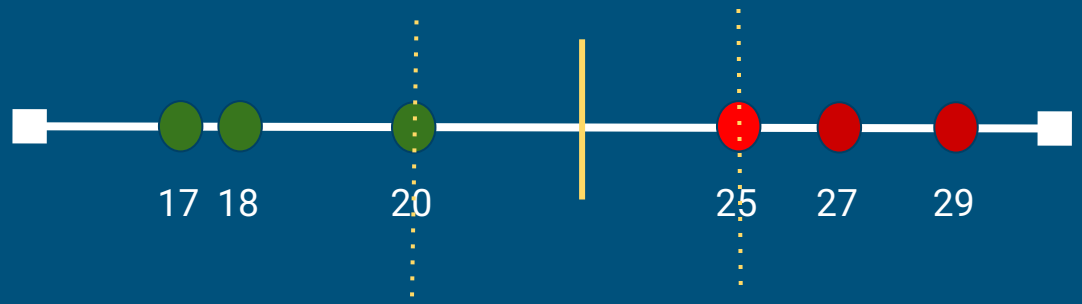


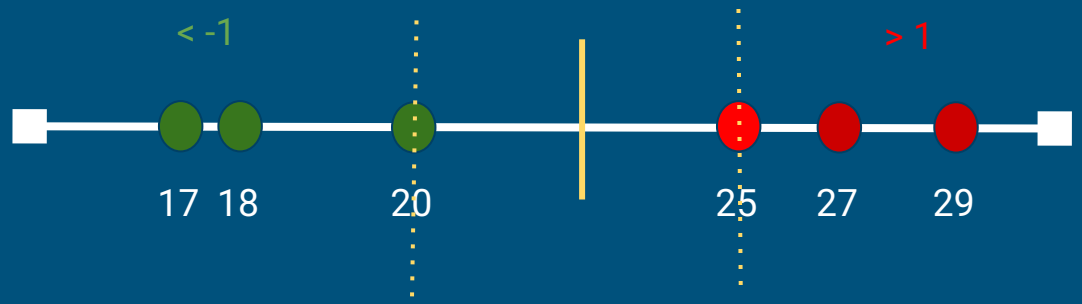
Choose the mid point of as the threshold - Maximum Margin Classifier

Maximum Margin Classifiers



Construct the widest road possible between the data points.
Median of the road is the threshold

Maximum Margin Classifiers



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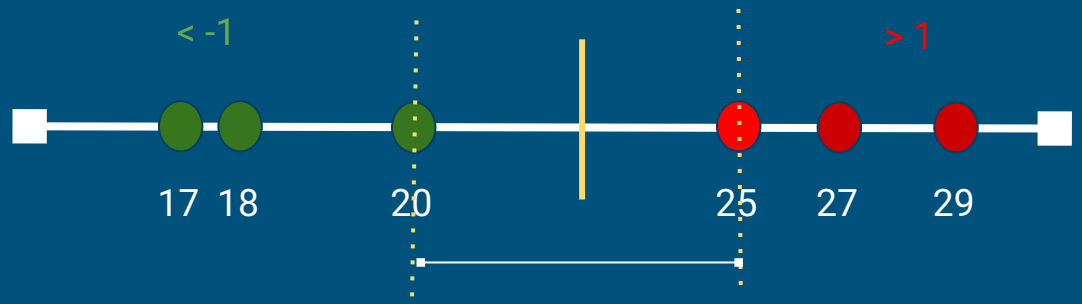
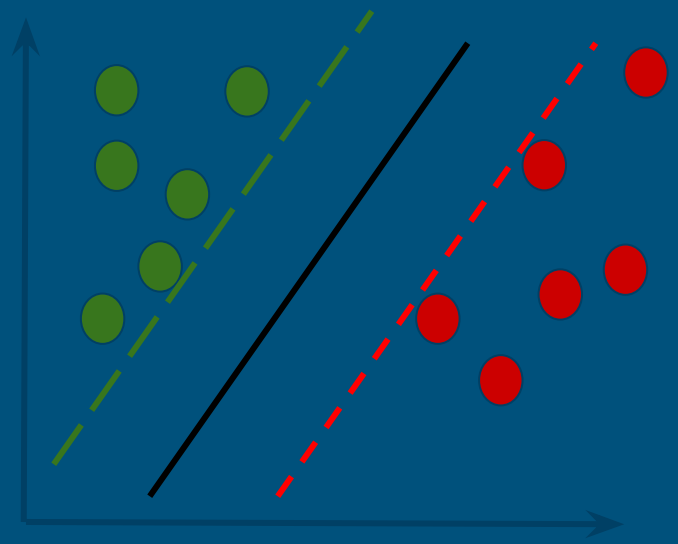
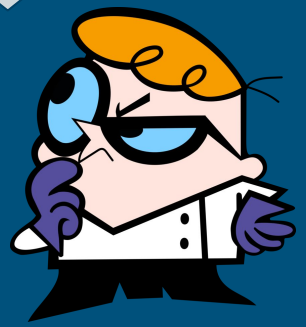


Maximise the width of the road

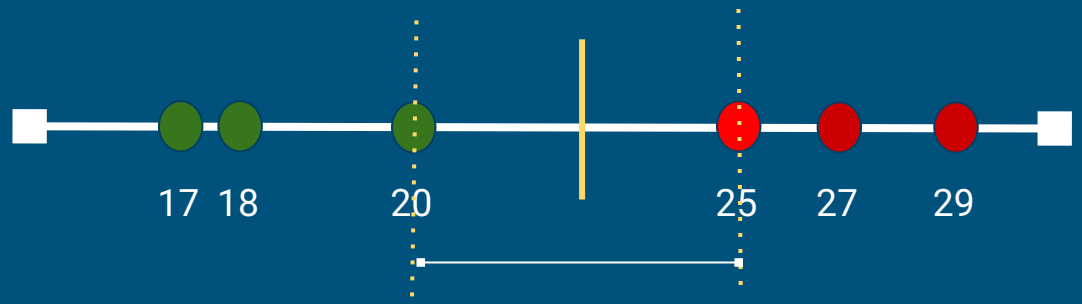
Maximum Margin Classifiers



What about 2-D?



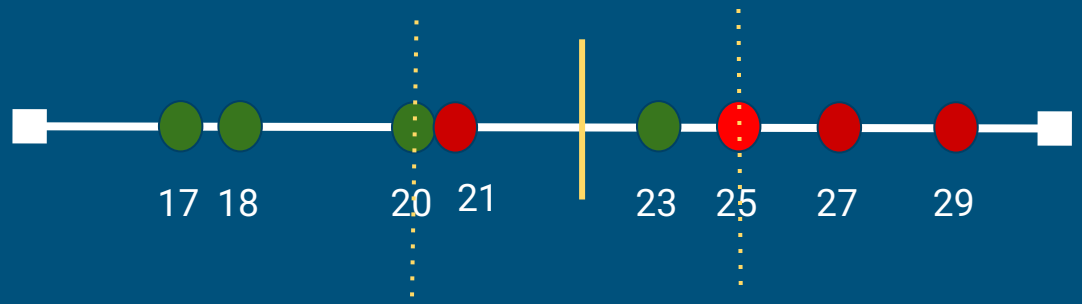
Maximum Margin Classifiers



Soft Margin Classifiers

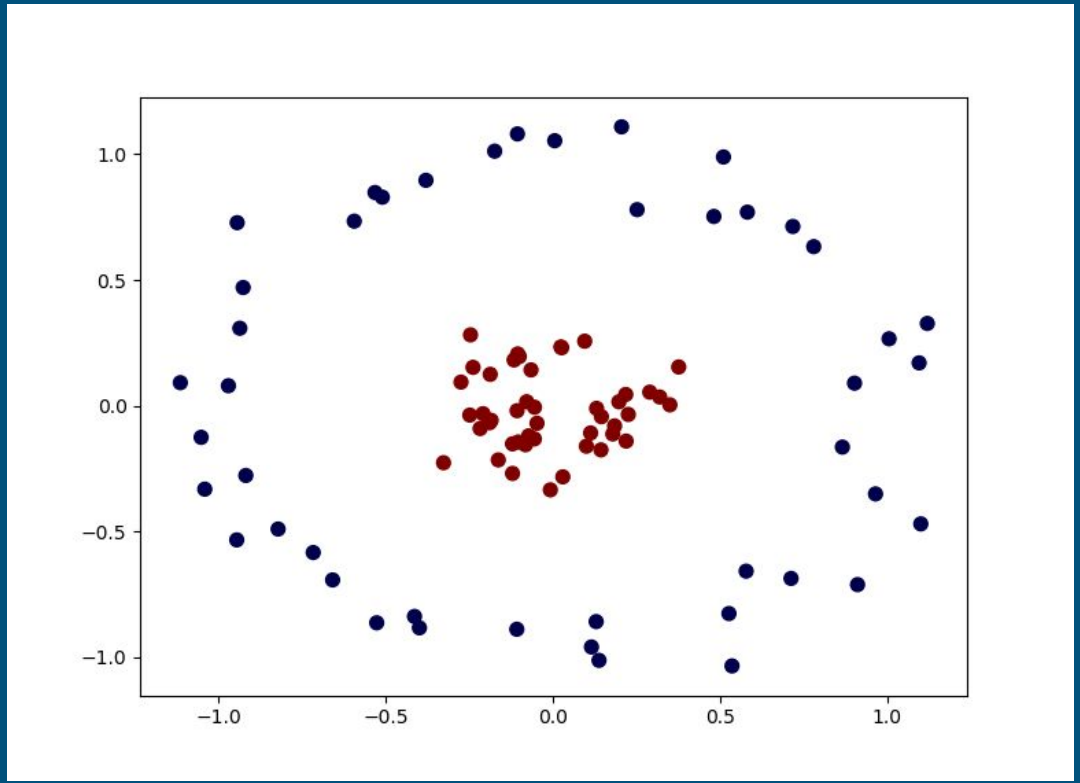
What to do in this case?

Soft Margin Classifiers



We allow for some errors. This is a **Soft Margin Classifier**

Linearly Non - Seperable data



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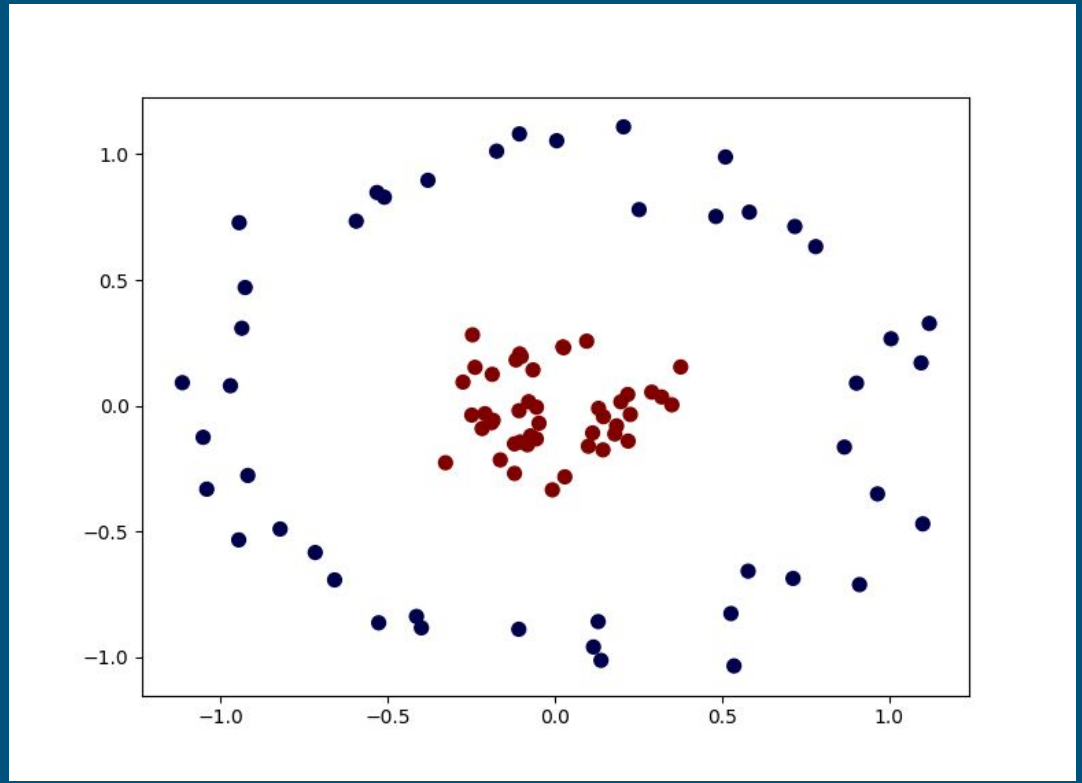

This problem shall be solved using Soft Margin Classifier in higher dimen-sons

Linearly Non - Seperable data

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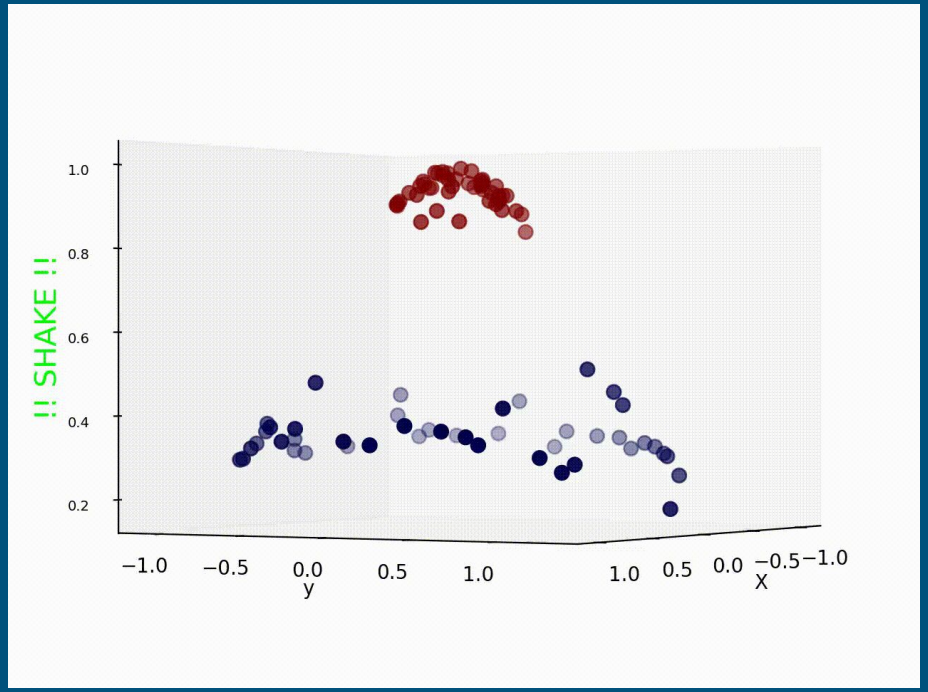
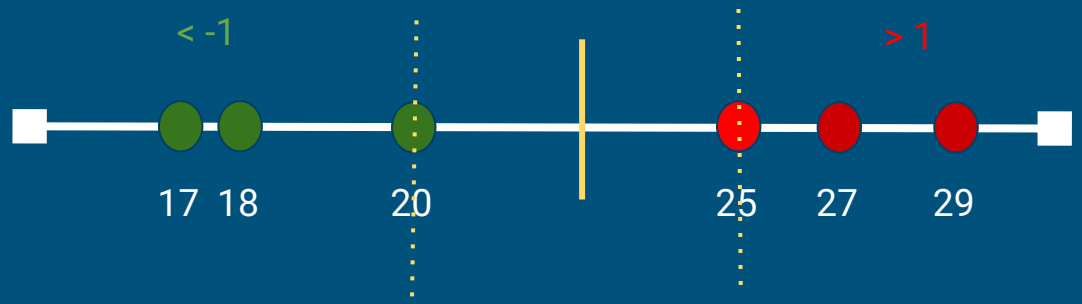


https://github.com/suvoooo/Machine_Learning/blob/master/SVMdemo.py

Math

Maximum Margin Classifiers



Construct the widest road possible between the data points.
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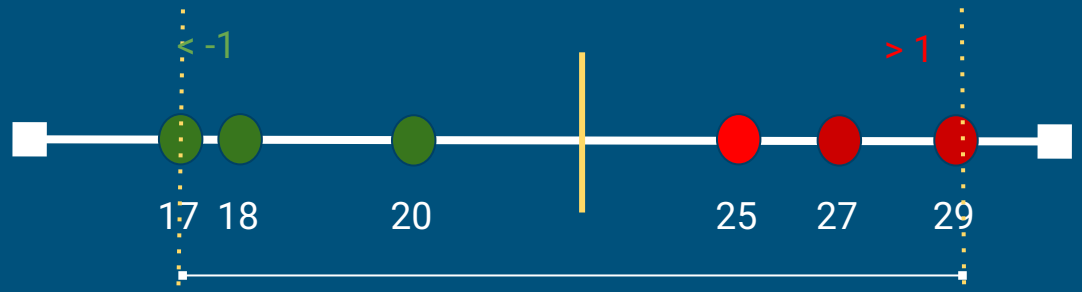
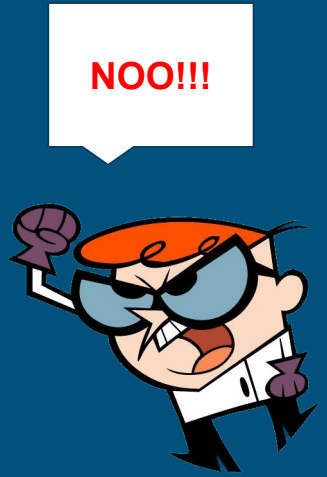
Maximum Margin Classifiers





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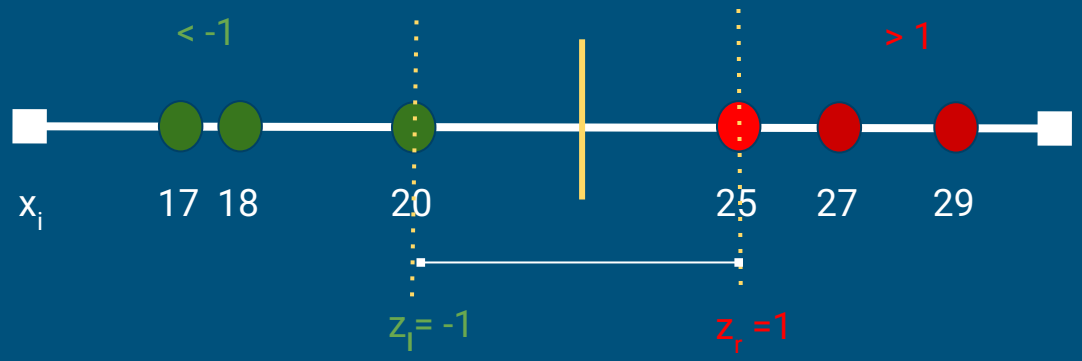
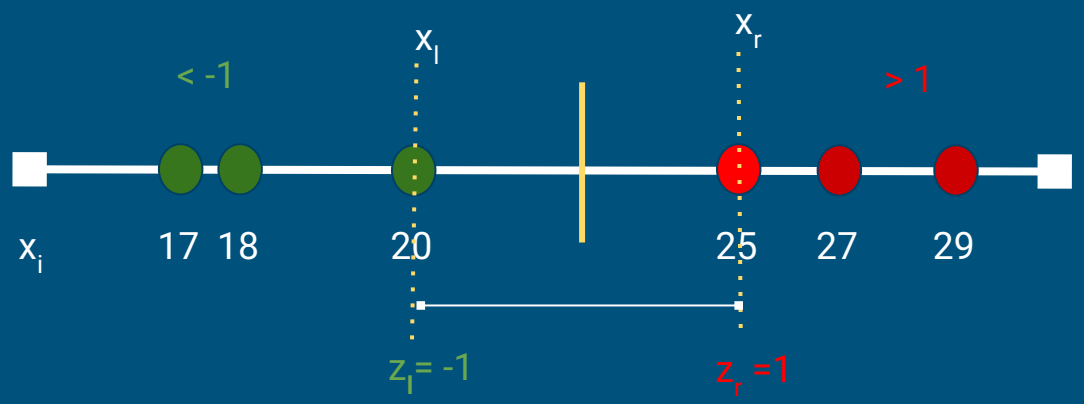


Maximise the width of the road

$w\, x_i + b \geq 1$, Obese

$w\, x_i + b \leq -1$, Not Obese

Maximum Margin Classifiers



$$\text{Width} = |x_r - x_l|$$
$$= \frac{z_r - b}{w} - \frac{z_l - b}{w}$$
$$= \frac{2}{|w|}$$

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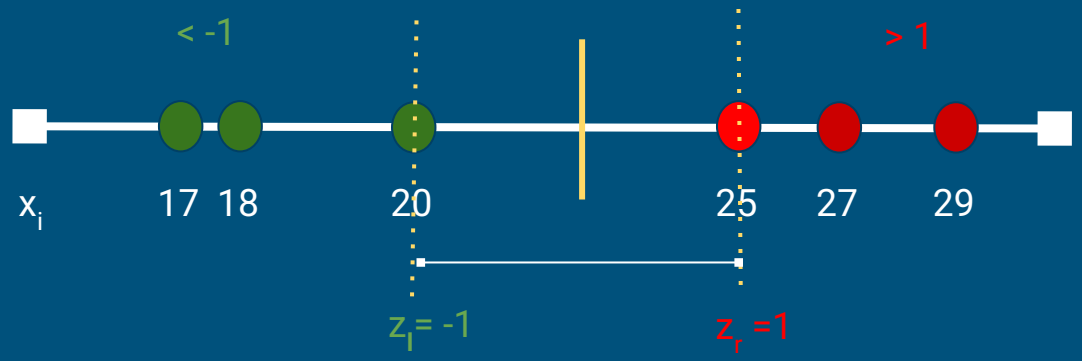

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Min $|w|$

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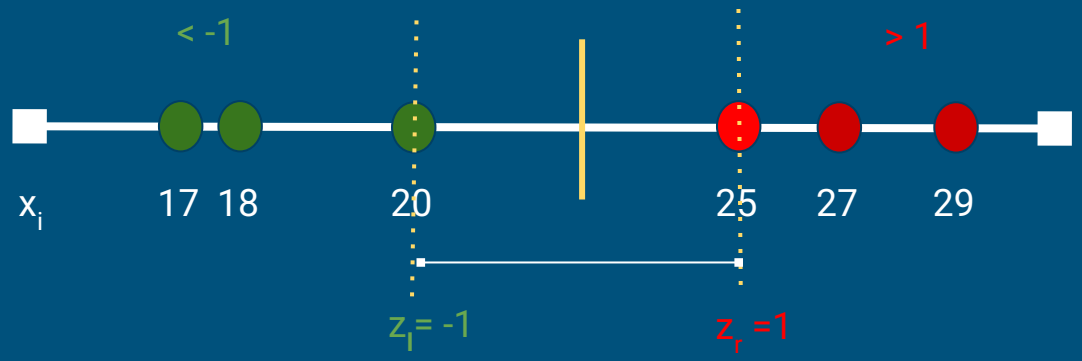
$$\text{Width} = |x_r - x_l| = \frac{z_r - b}{w} - \frac{z_l - b}{w} = \frac{2}{|w|}$$

Maximum Margin Classifiers



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$w\,x_i + b \leq -1$, Not Obese

Min $|w|^2/2$

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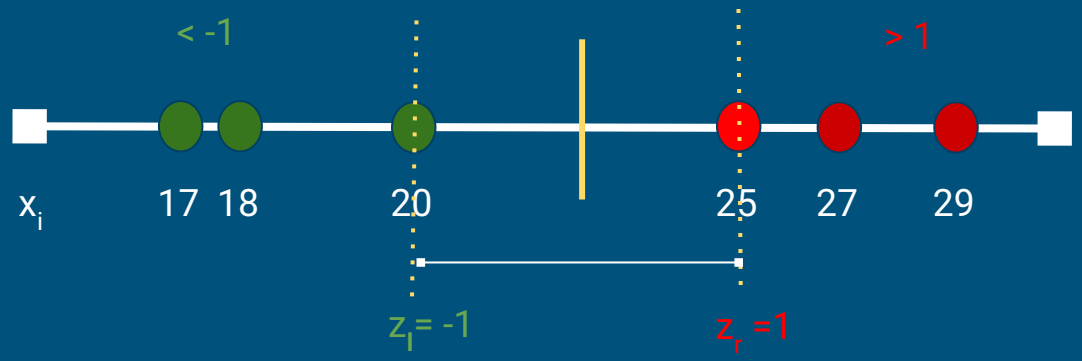
Maximum Margin Classifiers



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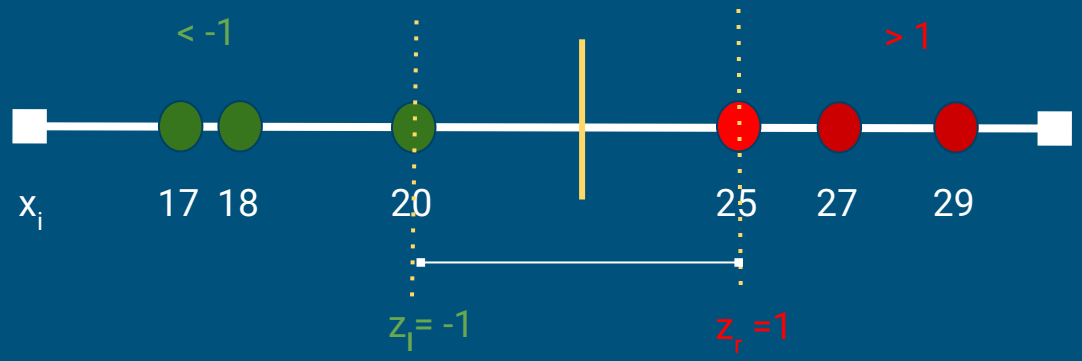
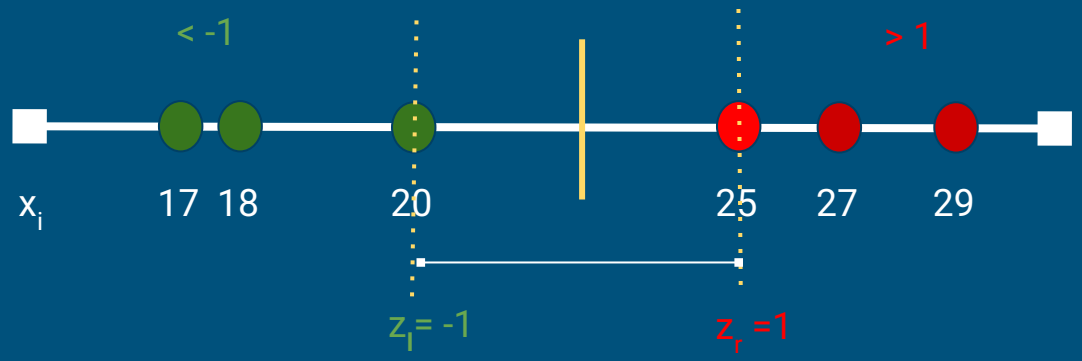


$y_i = 1$, Obese

$y_i = -1$, Not Obese

Maximum Margin Classifiers



y_i = 1, Obese

y_i = -1 , Not Obese

Min $|w|^2/2$

$w\, x_i + b \geq 1$, Obese

$w\, x_i + b \leq -1$, Not Obese

Min $|w|^2/2$

$y_i\, (w\, x_i + b) \geq 1$, Obese

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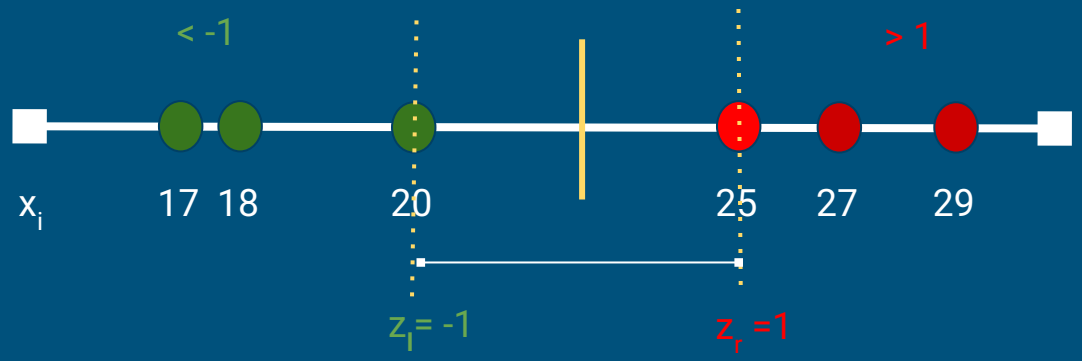
Maximum Margin Classifiers



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Min $|w|^2/2$

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Min $|w|^2/2$

$y_i\, (w\, x_i + b) - 1 \geq 0$

Maximum Margin Classifiers

$$\text{Min } |w|^2/2$$

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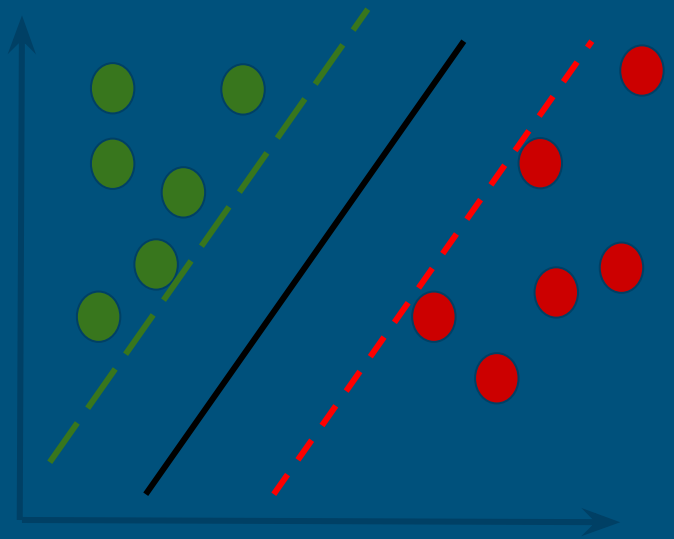

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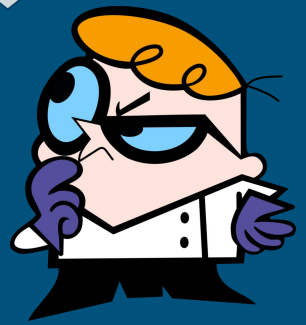

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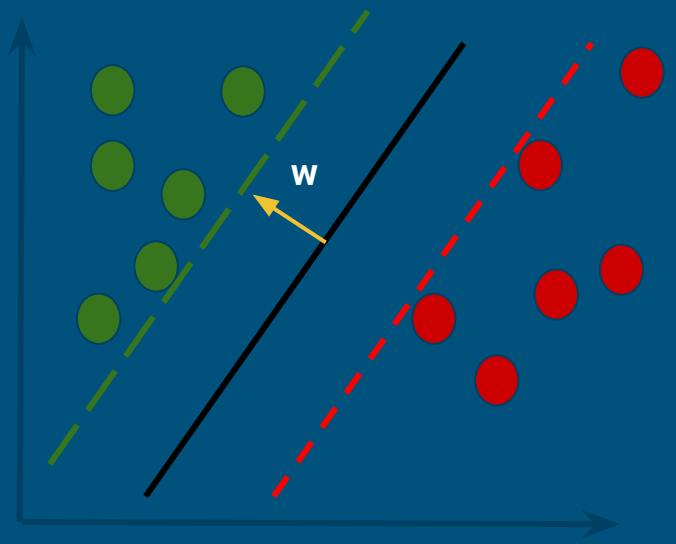


$$\text{Min } |w|^2/2$$

$$y_i\,(w\,x_i + b) - 1 \geq 0$$

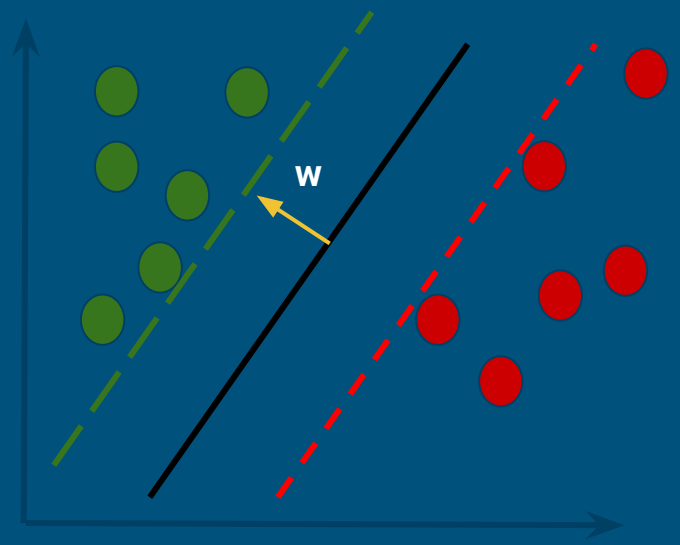
$$\text{Min } \|\mathbf{w}\|^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$

Maximum Margin Classifiers

Min $||\mathbf{w}||^2/2$

$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$



Maximum Margin Classifiers

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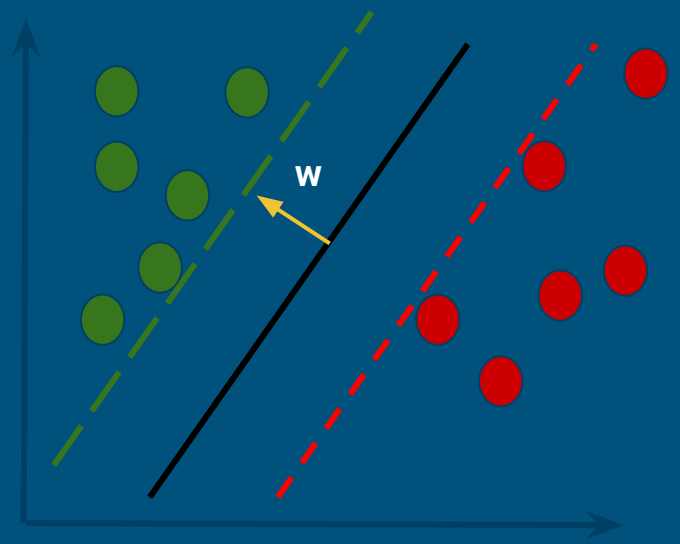
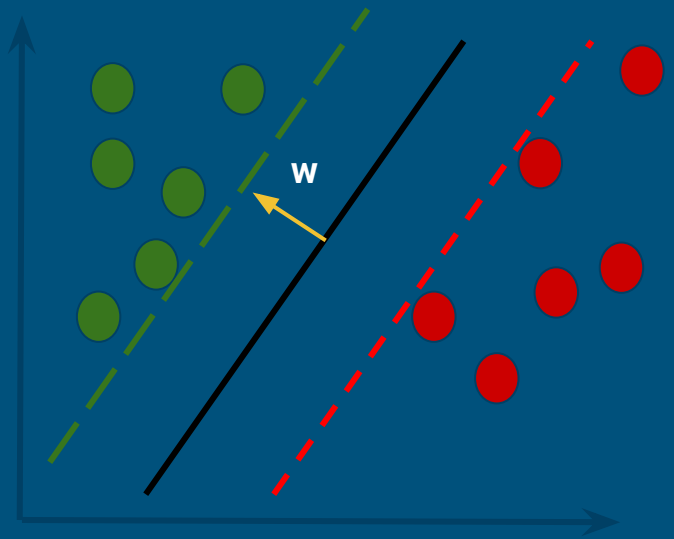


Now What?!?!

Maximum Margin Classifiers

Min $||\mathbf{w}||^2/2$

$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$

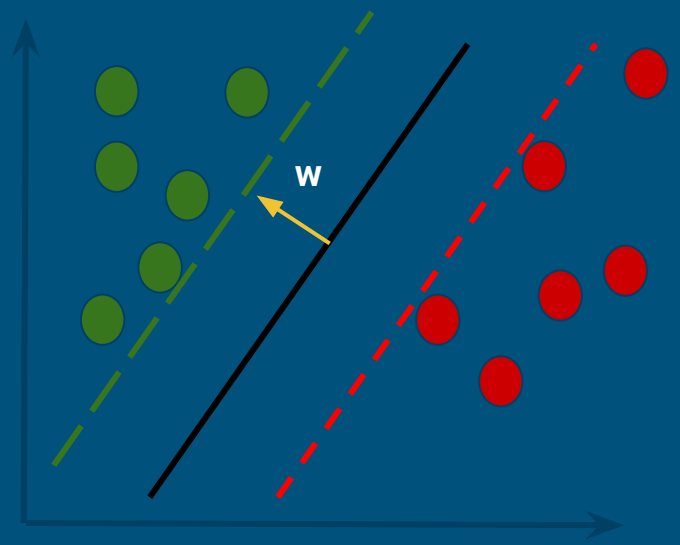


We shall use the method of Lagrange Multipliers to solve the problem

Maximum Margin Classifiers

$$\text{Min } \|\mathbf{w}\|^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$



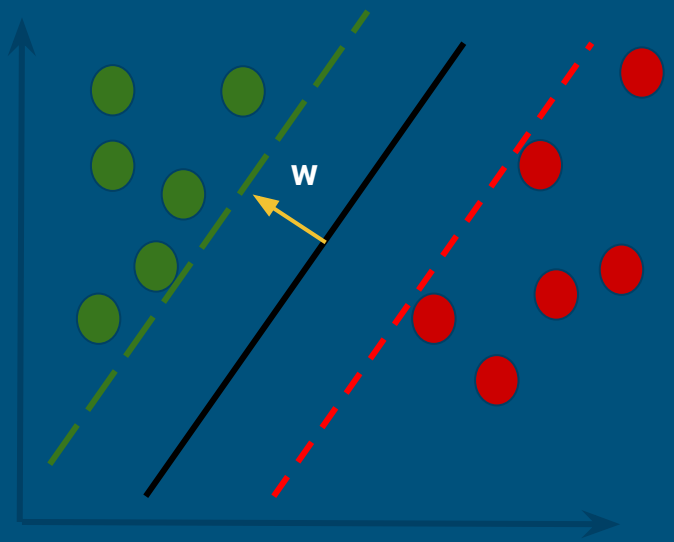
$$L(\mathbf{w}, b, \alpha_i) = \|\mathbf{w}\|^2/2 - \Sigma_i\, \alpha_i (y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1)$$

Lagrangian

Maximum Margin Classifiers

Min $||\mathbf{w}||^2/2$

$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$



$L(\mathbf{w}, b, \alpha_i) = ||\mathbf{w}||^2/2 - \Sigma_i\, \alpha_i (y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1)$

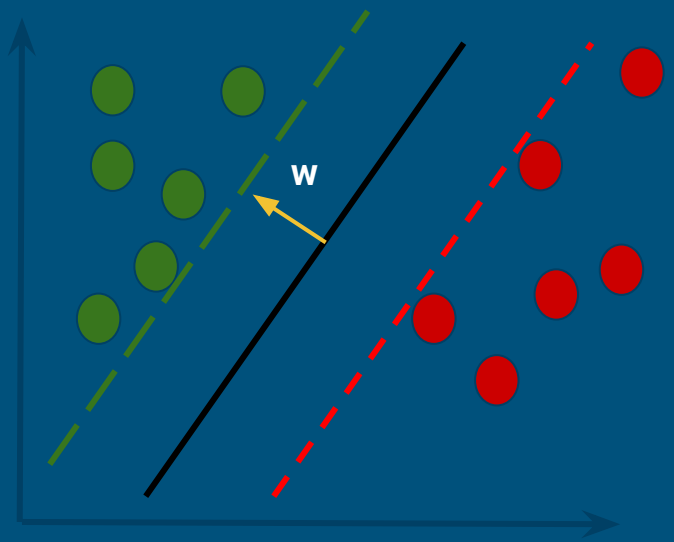
Lagrangian

Differentiate w.r.t **w** and **b** and set it to **zero**

Maximum Margin Classifiers

$$\text{Min } \|\mathbf{w}\|^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$



$$L(\mathbf{w}, b, \alpha_i) = \|\mathbf{w}\|^2/2 - \Sigma_i\, \alpha_i (y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1)$$

Lagrangian

$$dL/d\mathbf{w} = \mathbf{w} - \Sigma_i\, \alpha_i y_i\, \mathbf{x}_i = 0$$

$$\mathbf{w} = \Sigma_i\, \alpha_i y_i\, \mathbf{x}_i$$

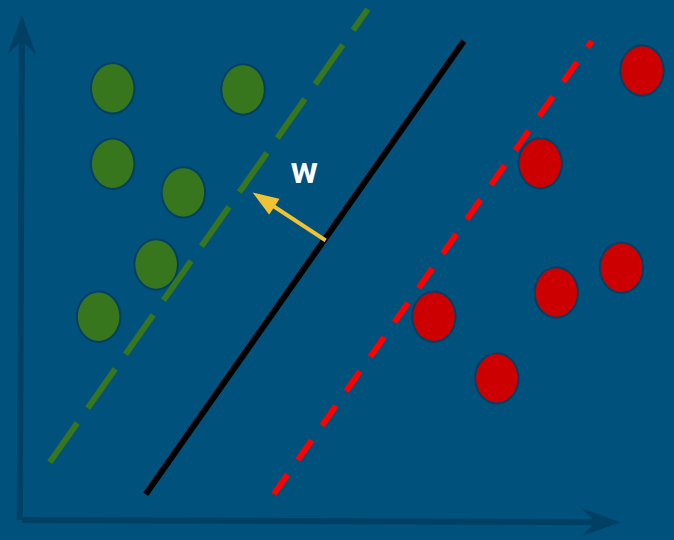
$$dL/db = -\Sigma_i\, \alpha_i y_i = 0$$

$$\Sigma_i\, \alpha_i y_i = 0$$

Maximum Margin Classifiers

$$\text{Min } \|\mathbf{w}\|^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$



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$$L(\alpha_i) = \Sigma_i\, \alpha_i - 1/2\, \Sigma_i \Sigma_i\, \alpha_j\, \alpha_i\, y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i$$

Maximum Margin Classifiers

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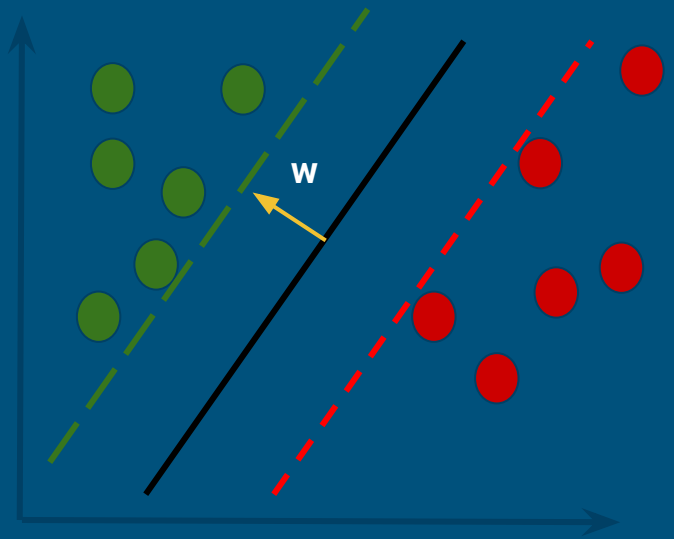
$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$



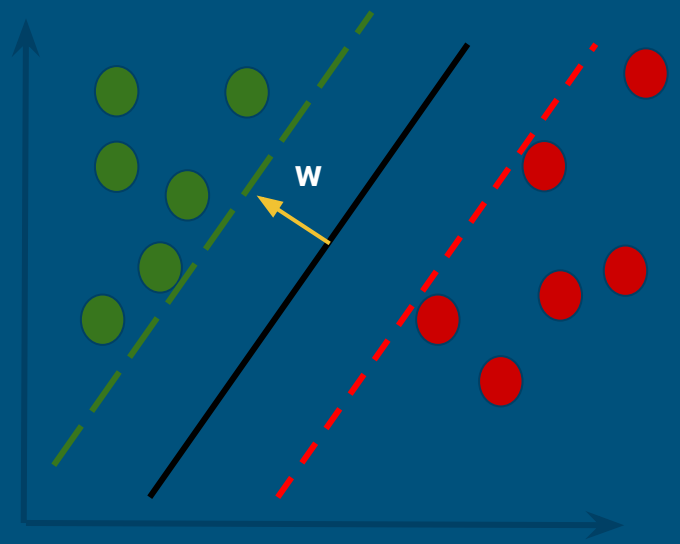
$$L(\alpha_i) = \Sigma_i\, \alpha_i - 1/2\, \Sigma_i \Sigma_i\, \alpha_j\, \alpha_i\, y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i$$

Maximize the lagrangian dual

Maximum Margin Classifiers

$$\text{Min } \|\mathbf{w}\|^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$



$$L(\alpha_i) = \Sigma_i\, \alpha_i - 1/2\, \Sigma_i \Sigma_i\, \alpha_j\, \alpha_i\, y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i$$

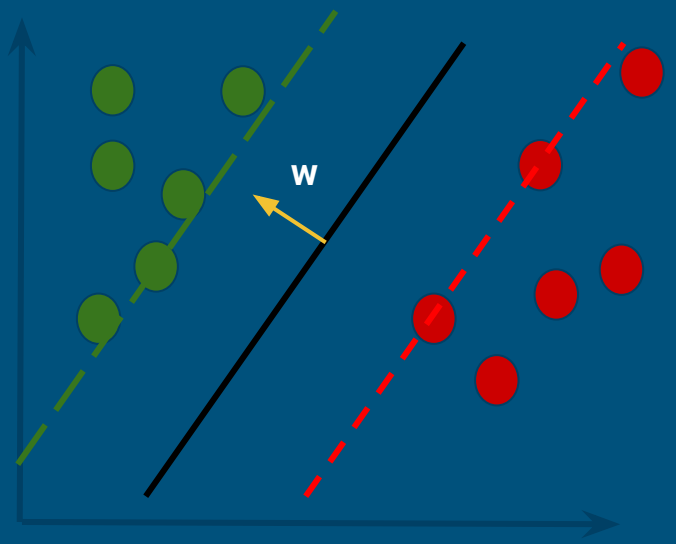
$$\max L(\alpha_i)$$

$$\Sigma_i\, \alpha_i y_i = 0\,,\ \alpha_i \geq 0$$

Maximum Margin Classifiers

$$\text{Min } \|\mathbf{w}\|^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$

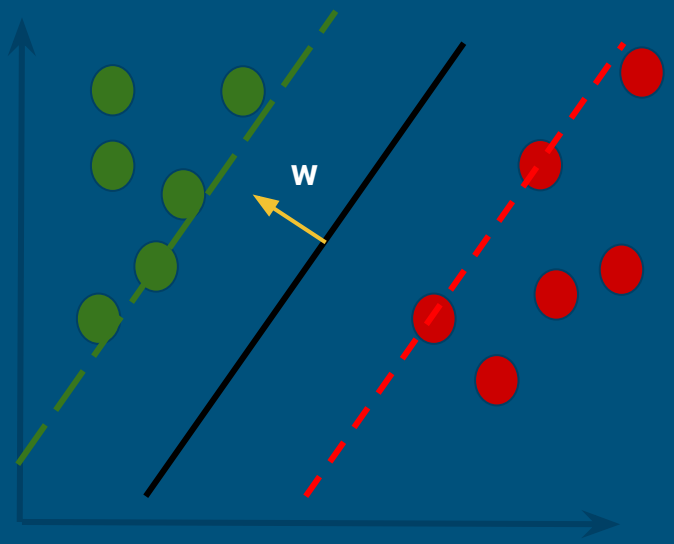


$$\max L(\alpha_i)$$

$$\Sigma_i\, \alpha_i y_i = 0\,,\ \alpha_i \geq 0$$

Standard Quadratic Program that can be solved using solvers

Maximum Margin Classifiers



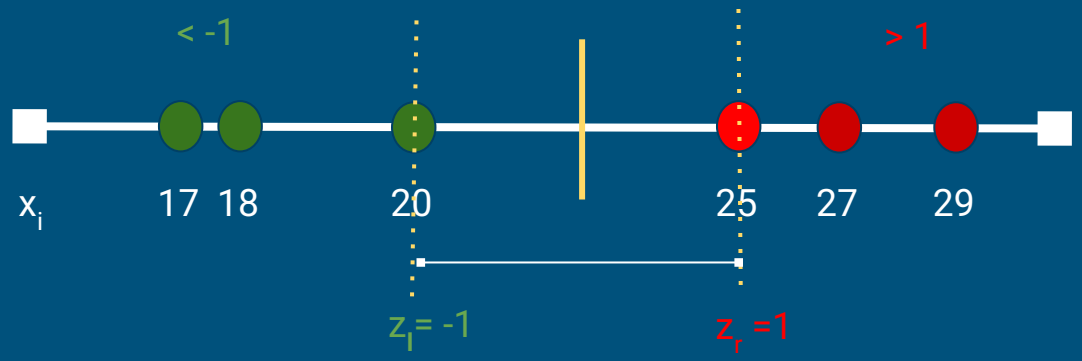
- $w = \Sigma \alpha_i y_i x_i$, $b = y_k - \mathbf{w} \cdot \mathbf{x_k}$ for any x_k such that $\alpha_k <> 0$
- Each non-zero α_i indicates that corresponding xi is a support vector.
- Then the classifying function will have the form:

$$f(\mathbf{x}) = \Sigma \alpha_i y_i \mathbf{x_i} \cdot \mathbf{u} + b$$

- $L(\alpha_i) = \Sigma_i \alpha_i - 1/2 \Sigma_i \Sigma_i \alpha_j \alpha_i y_j y_i \mathbf{x_j} \cdot \mathbf{x_i}$

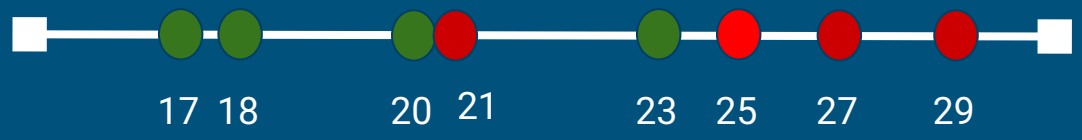
Both the optimisation and decision functions depend only on dot products

Soft Margin Classifiers



Solved

Soft Margin Classifiers

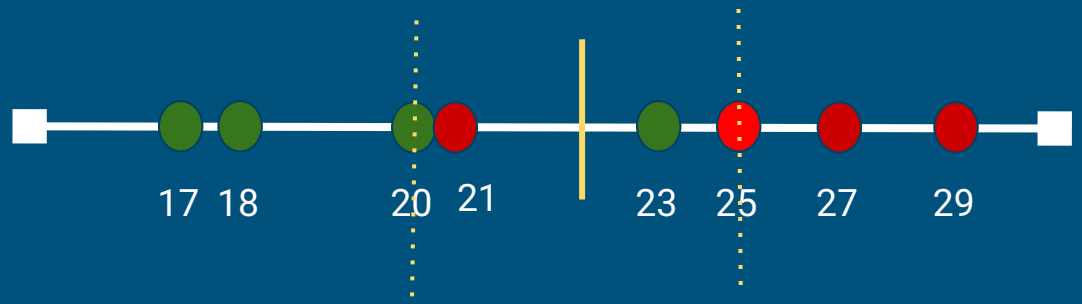


What to do in this case?

Soft Margin Classifiers

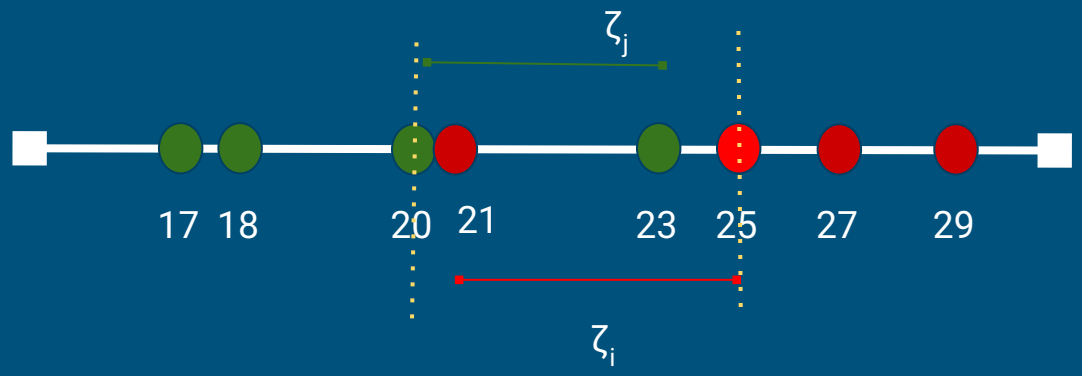
What to do in this case?

Soft Margin Classifiers



We allow for some errors. This is a **Soft Margin Classifier**

Soft Margin Classifiers



$$\text{Min } ||\mathbf{w}||^2/2$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1 \geq 0$$

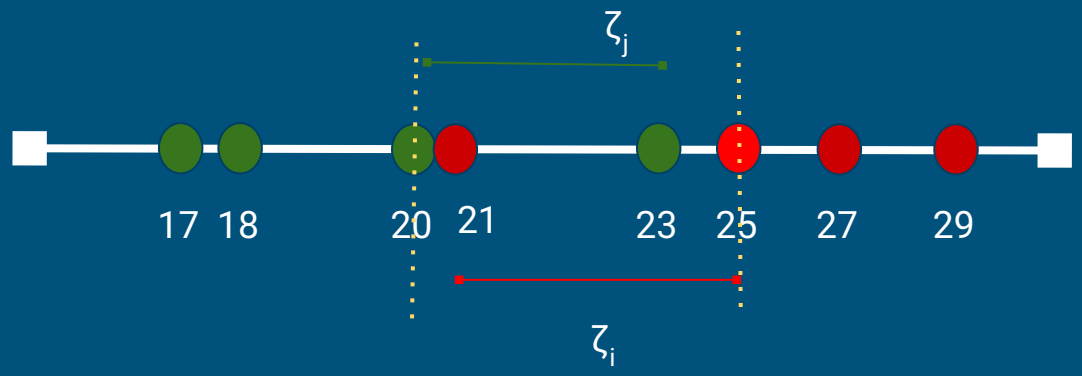
$$\text{Min } ||\mathbf{w}||^2/2 - C\,\Sigma\zeta_i$$

$$y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \zeta_i$$

$$\zeta_i \geq 0$$

$$\zeta_i = \max(0, y_i\,(\mathbf{w} \cdot \mathbf{x}_i + b) - 1)$$

Soft Margin Classifiers



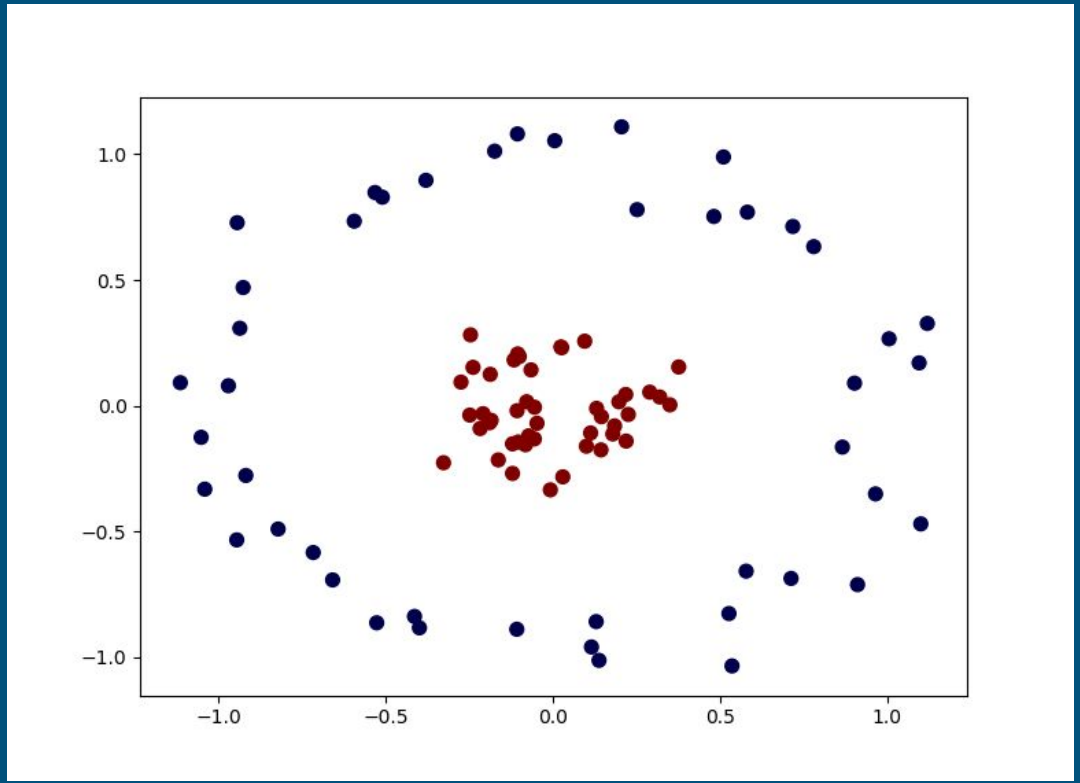
$L(\alpha_i) = \Sigma_i \alpha_i - 1/2 \Sigma_i \Sigma_i \alpha_j \alpha_i y_j y_i \mathbf{x}_j \cdot \mathbf{x}_i$

- $b = y_k(1 - \zeta_k) - \mathbf{w} \cdot \mathbf{x}_k$

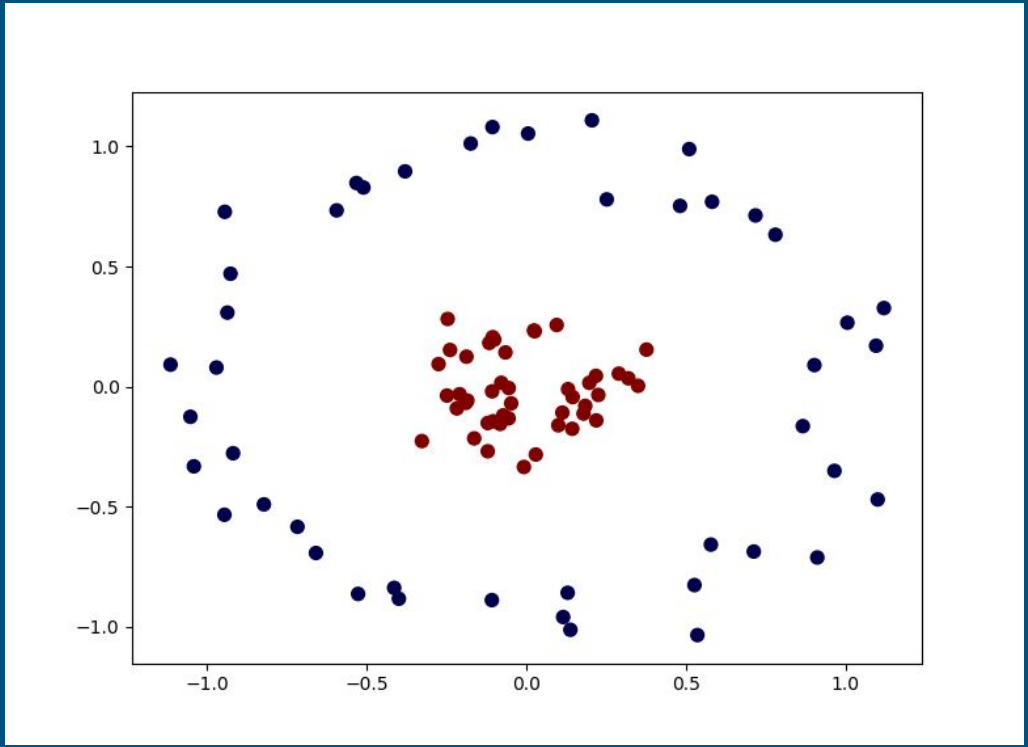
$f(\mathbf{x}) = \Sigma \alpha_i y_i \mathbf{x}_i \cdot \mathbf{u} + b$

Both the optimisation and decision functions depend only on dot products

Linearly Non - Seperable data



Linearly Non - Seperable data



$$\text{Min } \|\mathbf{w}\|^2/2 - C\,\Sigma\zeta_i$$

$$y_i\,(\mathbf{w} \cdot \varphi(\mathbf{x_i}) + b) \geq 1 - \zeta_i$$

$$\zeta_i \geq 0$$

$w = \Sigma\alpha_i y_i \varphi(\mathbf{x_i})$, $b = y_k - \mathbf{w} \cdot \varphi(\mathbf{x_k})$ for any x_k such that $\alpha_k <> 0$

$$f(\mathbf{u}) = \Sigma\alpha_i y_i \varphi(\mathbf{x_i}) \cdot \varphi(\mathbf{u}) + b$$

$$L(\alpha_i) = \Sigma_i\, \alpha_i - 1/2\, \Sigma_i \Sigma_j\, \alpha_j\, \alpha_i\, y_j y_i \varphi(\mathbf{x_i}) \cdot \varphi(\mathbf{x_j})$$

Linearly Non - Seperable data

Both the optimisation and decision functions depend only on dot products

$$\text{Min } \|\mathbf{w}\|^2/2 - C\,\Sigma\zeta_i$$

$$y_i\,(\mathbf{w} \cdot \varphi(\mathbf{x_i}) + b) \geq 1 - \zeta_i$$

$$\zeta_i \geq 0$$

$w = \Sigma\alpha_i y_i \varphi(\mathbf{x_i})$, $b = y_k - \mathbf{w} \cdot \varphi(\mathbf{x_k})$ for any x_k such that $\alpha_k <> 0$

$$f(\mathbf{u}) = \Sigma\alpha_i y_i \varphi(\mathbf{x_i}) \cdot \varphi(\mathbf{u}) + b$$

$$L(\alpha_i) = \Sigma_i\,\alpha_i - 1/2\,\Sigma_i\Sigma_j\,\alpha_j\,\alpha_i\,y_j y_i \varphi(\mathbf{x_i}) \cdot \varphi(\mathbf{x_j})$$

Kernel Trick

Function K such that

$K(\mathbf{x}_i, \mathbf{x}_j) = \varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}_j)$

Kernel Trick

Function K such that

$K(\mathbf{x}_i,\mathbf{x}_j) = \varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}_j)$

So, **Kernel function defines inner product in the transformed space**

Kernel Trick

Function K such that

$K(\mathbf{x}_i, \mathbf{x}_j) = \varphi(\mathbf{x}_i) \cdot \varphi(\mathbf{x}_j)$

- Polynomials of degree exactly *d*

$$K(\mathbf{u}, \mathbf{v}) = (\mathbf{u} \cdot \mathbf{v})^d$$

- Polynomials of degree up to *d*

$$K(\mathbf{u}, \mathbf{v}) = (\mathbf{u} \cdot \mathbf{v} + 1)^d$$

- Gaussian kernels

$$K(\vec{u}, \vec{v}) = \exp\left(-\frac{||\vec{u} - \vec{v}||_2^2}{2\sigma^2}\right)$$

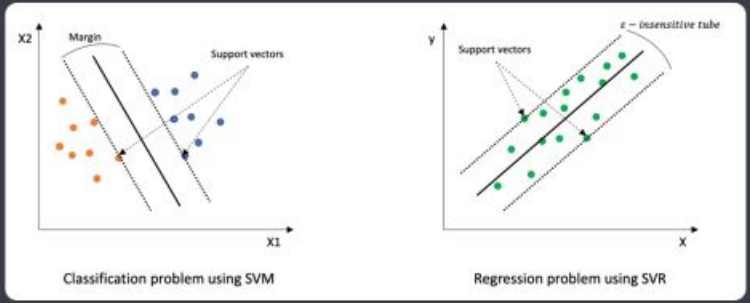
- Sigmoid

$$K(\mathbf{u}, \mathbf{v}) = \tanh(\eta \mathbf{u} \cdot \mathbf{v} + \nu)$$

- And many others: very active area of research!

Take Aways

- Maximum Margin Classifier
- Soft MArgin Classifier
- Mapping data points from low dimensional space to a higher dimensional space can make it possible to apply SVM even for non-linear data sample.
- We don’t need to know the mapping function itself, as long as we know the Kernel function ; ***Kernel Trick***

Introduction to Support Vector Regression (SVR)

- Support Vector Regression (SVR) extends SVM principles from classification to regression problems
- Goal: Predict continuous values rather than discrete class labels
- Key idea: Find a function that deviates from the actual target values by a value no greater than ε for each training point